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Introduction to Data Mining

Data mining: extracting useful patterns and knowledge from large datasets using various techniques, with the aim of uncovering or unmining the hidden patterns, correlations, anomalies, or trends that can be used to make informed decisions.

- **Healthcare Example: Disease Prediction**
  - **Scenario**: Predicting the risk of diabetes based on patient data.
  - **Data**: Patient records with attributes like age (X1), weight (X2), blood sugar levels (X3), and family medical history (X4).
  - **Analysis**: Using a logistic regression model:
    - Hypothesize a model: $\text{Risk of Diabetes} = \beta_0 + \beta_1 \cdot X1 + \beta_2 \cdot X2 + \beta_3 \cdot X3 + \beta_4 \cdot X4$
    - Estimate coefficients β_i using training data.
    - Predict the probability of diabetes for new patients based on their attributes.
  - **Insight**: The model predicts the likelihood of diabetes for patients, allowing healthcare providers to prioritize interventions for high-risk individuals.

Why do we need data mining?
- identifying patterns and trends based on data
- making predictions based on historical data
- discovering relationships and correlation between variables
- detecting anomalies and outliers

predictive : predicting the future
prescriptive : no predictions

Steps involved in data mining

1. Problem definition: clearly define the problem and objectives to be achieved.
   E.g. a hospital wants to predict which patients are likely to be readmitted within 30 days of discharge.
2. Data collection: gather relevant data from various sources. We need to ensure that the data is accurate, complete, and reliable.
   E.g. demographics (a patient in a hilly, colder region could be more likely to be readmitted), medical history, treatment received, lab results, discharge summaries, etc.
3. Data preprocessing: prepare the data for analysis.
   Data cleaning: handling missing values, remove duplicates, correcting errors.
   Data integration: combining data from multiple sources.
   Data transformation: normalize, discretize, or analyze.
4. Model building: selecting the appropriate data mining technique.
   Includes splitting the data into training and testing sets. Model building techniques include classification, regression, CNN, etc.
5. Model evaluation: assess the model's performance using various metrics so that it validates for unseen data.
   Metrics: accuracy, precision, recall, F1 score, etc.
   Validation techniques: cross validation, confusion matrix, etc.
   E.g. We have a binary classifier that tells if a patient is readmitted after 30 days of discharge.

confusion matrix

actual \ predicted (A/NA)	A	NA
A	TP	FN
NA	FP	TN

$$\text{Accuracy} = \frac{\phi(TP) + \phi(TN)}{\Sigma \text{all}}$$ ← proportion of predictions that are correct

$$\text{Precision} = \frac{\phi(TP)}{\phi(TP) + \phi(FP)}$$ ← proportion of tve predictions that are correct.

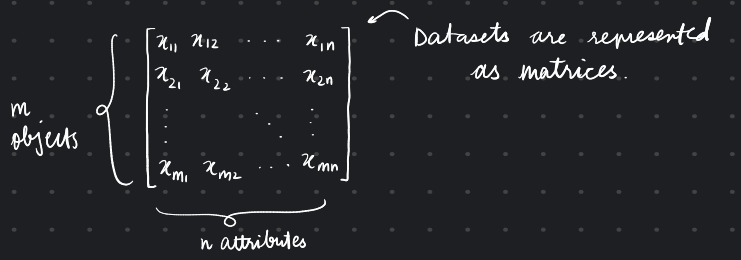
$$\text{Recall} = \frac{\phi(TP)}{\phi(TP) + \phi(FN)}$$ ← proportion of actual tves that are predicted as tves correctly.

$$\text{F1-score} = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

6. Deployment: deploy the model in the real world to make predictions or provide insights and also to monitor the model's performance over time.
7. Result interpretation: analyze the results of the findings and communicate those to the stakeholders.
   E.g. to indentify factors influencing patient readmissions and to prepare strategies to reduce the readmission rate.

	Data mining	Data warehousing	Data analytics
Purpose	- To discover hidden patterns and relationships in data.	- To provide a centralized repository for integrated data from multiple sources.	- To analyze the data for insights and informed decisions.
Process	- Involves various algos and techniques to extract patterns. E.g. classification, regression, NN, etc.	- Involves data integration, storage, management.	- Involves statistical analysis, data modeling, data visualization, predictive analytics on the data.
Output	- Patterns, pre-trained model, rules, etc.	- Consolidated data that is ready for analysis.	- Predictions and recommendations.

Datasets, objects and attributes:
In general, we will be talking about data sets (collections of information). These datasets are made up of data objects.
E.g. A medical database: a record for one patient could be an object.
Each data object will be having some properties. These are called attributes (data fields/ features/ variables).



E.g.

Patient ID	Name	Age	Gender	Blood Pressure	Diagnosis	Admission Date	Height (cm)	Weight (kg)
1	Alice Smith	45	Female	120/80	Hypertension	2023-01-15	165	68
2	Bob Johnson	56	Male	140/90	Diabetes	2023-01-20	175	85
3	Charlie Brown	35	Male	110/70	No known condition	2023-01-25	180	75
4	Diana Prince	29	Female	115/75	Anemia	2023-01-30	170	60

- each patient is an object.

Data types

1. Based on the attributes:

i. Categorical data (quantitative data): It represents categories or groups, but at the same time it does not have any numeric meaningful representation.

Subcategories:
- Nominal data:
  nominal -> related to names
  It is what we store in "enum" types. Each value represents some kind of category, code, or state.
  E.g. Hair color: {black, white, grey, silver, blonde}. It takes from a pre-defined set of values.
  Properties:
  - Categories are mutually exclusive (hair color won't be black and blonde at the same time).
  - Typically represented by names/ labels.
  - No inherit ranking or ordering in categories (black and blonde can't be ranked in a sensible manner).
  Binary (or boolean) attributes are a specific case of nominal attributes that contain 2 classes (0/ 1).
  E.g. diabetic or not, married or not (here we are considering the patients dataset).
  (unequal weights, asymmetric, will affect other considerations), (equal weights, symmetric)

  In general, the more important (rarer) attribute is denoted by '1' or True.
- Ordinal data:
  It consists of categorical data with some meaningful ranking or ordering.
  E.g. Qualification: {high school, graduate, post graduate, PhD},
  2. Customer rating: {poor, fair, good, excellent}.
  Properties:
  - Mutually exclusive categories.
  - Inherent ordering.
  - Differences between categories are not uniform quantifiable (can't tell how greater a PhD holder is over a post graduate).

ii. Numerical data (qualitative data): It represents measurable quantities that can be represented in #s.

Subcategories:
- Discrete data:
  It consists of distinct and separate values that can be counted. Generally, integers come under this category, which represent the count of items/ occurrences.
  E.g. cars in a parking lot, # of students in a class, ids of patients, etc.
  Can be countably finite or infinite.
  countably infinite values means that they can be put in one-to-one correspondence with the set of natural #s
  Properties:
  - Can only take countable values.
  - No intermediate values exist between 2 adjacent values.
- Continuous data:
  It takes values in a given range. These values are often through measurements and can take infinite possible values within the specified range.
  E.g. body temperature.

Other subcategories of numerical data:
Interval scaled and ratio scaled attributes:

Interval scaled
- numerical attributes where difference between values is meaningful but there's no true zero point.
  0 doesn't indicate absence of attribute value, 0 is a valid attribute value itself.
- E.g. temperature (0°C doesn't signify absence of temp).
  Diff. is meaningful: 40°C - 20°C is same as 60°C - 40°C
  Scale doesn't start with 0: 40°C is not twice of 20°C

Ratio scaled
- interval values + true 0 point.
  (difference is meaningful; 0 signifies absence of attribute values.)
- E.g. height of a person (0cm height would indicate absence of value).
  Diff is meaningful: 40cm - 20cm = 100cm - 80cm
  scaled: 120cm is twice as tall as 60cm

Q. Identify the data type of each attribute:

Patient ID	Name	Age	Gender	Blood Pressure	Diagnosis	Admission Date	Height (cm)	Weight (kg)
1	Alice Smith	45	Female	120/80	Hypertension	2023-01-15	165	68
2	Bob Johnson	56	Male	140/90	Diabetes	2023-01-20	175	85
3	Charlie Brown	35	Male	110/70	No known condition	2023-01-25	180	75
4	Diana Prince	29	Female	115/75	Anemia	2023-01-30	170	60

numeric discrete

categorical nominal

numerical discrete ratio-scaled

categorical, nominal, binary (symmetric)

numeric [at the same time, could be categorized]

categorical, nominal

numerical, interval-scaled

here the data seems discrete, but we'll go with continuous and ratio scaled

Student ID	Name	Age	Gender	GPA	Major	Enrollment Date	Credits Earned	Attendance (%)
1	John Doe	20	Male	3.5	Computer Science	2021-09-01	45	95
2	Jane Smith	22	Female	3.8	Mathematics	2020-09-01	90	85
3	Emily Davis	21	Female	3.2	Biology	2019-09-01	75	90
4	Michael Brown	23	Male	3.9	Physics	2018-09-01	120	80

discrete

nominal

numerical (ratio-scaled) discrete

nominal

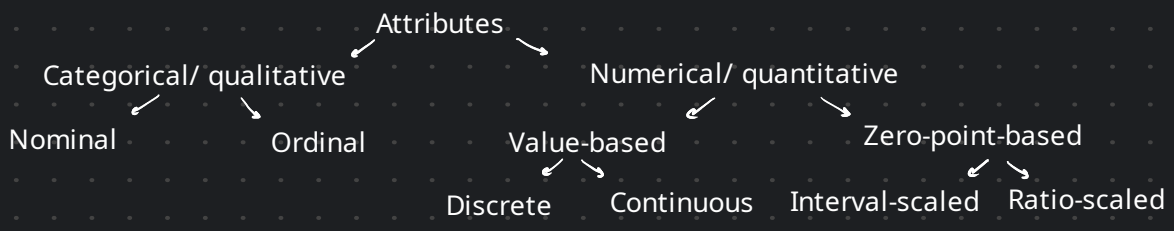
ratio-scaled continuous

nominal

numerical interval-scaled

ratio-scaled discrete

numerical, ratio-scaled,



2. Based on the structure:

i. Structured data: When data is organized in a highly predictable and pre-defined way, it is called structured.
- Organized data: each record/object will be storing fixed attributes.
- Querying is easier (SQL can be used for querying).

E.g.

ID	Name	Age	Gender	City
1	John Doe	30	Male	New York
2	Jane Smith	25	Female	Los Angeles
3	Bob Brown	35	Male	Chicago

5 fixed attributes
3 records

ii. Unstructured data: It lacks a pre-defined data model (data that doesn't fit under traditional databases).
- Unorganized data: can't be structured easily (we need to find some way to convert these to some semi-structured form).
- Hard to query/ not easily searchable.

E.g. text documents, multimedia.

Consider an email (could include attachments as well, unstructured data can be heterogeneous):

Subject: Regarding the Project Deadline

Hi team,

I hope this message finds you well. I wanted to discuss the upcoming project deadline. We need to finalize the design by the end of this week and start preparing the presentation materials. Please find the attached project proposal document for your review.

Best regards,
John

We could try and convert it to some sort of structured manner.

Attribute	Value
Subject	Regarding the Project Deadline
Sender	John
Recipients	Team
Body	Hi team, I hope this message finds you well. I wanted to discuss the upcoming project deadline. We need to finalize the design by the end of this week and start preparing the presentation materials. Please find the attached project proposal document for your review. Best regards, John
Attachments	Project Proposal Document (File)

iii. Semi-structured data: It falls in between structured and unstructured data, where it doesn't conform to a strict schema, but has some organizational properties.
- They contain some sort of heirarchical structure.
- Flexible schema with nested elements.
- Can be parsed and queried using specialized tools.
- Difficult indexing.

E.g. a JSON object, XML files.

```json
{
  "student": {
    "id": "101",
    "name": "Alice",
    "age": 22,
    "courses": ["Math", "Physics"],
    "address": {
      "city": "Boston",
      "zipcode": "02101"
    }
  }
}
```

```xml
<bookstore>
  <book category="fiction">
    <title>The Great Gatsby</title>
    <author>F. Scott Fitzgerald</author>
    <year>1925</year>
    <price>10.99</price>
  </book>
  <book category="non-fiction">
    <title>Thinking, Fast and Slow</title>
    <author>Daniel Kahneman</author>
    <year>2011</year>
    <price>15.99</price>
  </book>
</bookstore>
```

Q. Consider this data for a patient and answer the following questions:

Patient ID	Patient Name	Age	Diagnosis	Medical History	Notes
101	John Doe	45	Diabetes	Family history of heart disease	Patient needs regular check-ups
102	Jane Smith	32	Asthma	None	Allergic to penicillin
103	Michael Brown	50	Hypertension	Smoker	Recent knee surgery

Which type of data is the 'Age' column in the patient data table?

- A) Structured data
- B) Unstructured data
- C) Semi-structured data
- D) No specific type

2. **What type of data is represented by the 'Medical History' column in the patient data table?**

- A) Structured data
- B) Unstructured data
- C) Semi-structured data
- D) No specific type

NOTE: patient name has schema, first name: string, last name: string; It is structured.

3. Based on the time and space:

i. Temporal data: It is data associated with some timestamp. It is recorded when some event occurs/ measurement is taken.
- Time dependent, so we need to store timestamp along with attribute.
- Purpose: to capture change over time.
E.g. stock market data: the price of a specific stock varying over some time period.

Stock Symbol	Time	Open Price (USD)	High Price (USD)	Low Price (USD)	Close Price (USD)
AAPL	09:30	175.00	175.30	174.80	175.15
AAPL	10:30	175.15	176.00	175.00	175.80
AAPL	11:30	175.80	177.00	175.50	176.50
AAPL	12:30	176.50	177.20	176.00	176.80

Or temperature varying over time.

ii. Spatial data: It is data that identifies the geographic location and shape of objects on the Earth's surface.
E.g. GIS (Geographic Information System) data

ID	Type	Name	Coordinates	Attribute
1	Point of Interest	Empire State Building	(40.748817, -73.985428)	Landmark
2	Street Address	123 Market St	(37.774929, -122.419418)	Residential
3	Building Footprint	Office Building	[(37.775, -122.419), (37.774, -122.418)]	Commercial
4	Land Use Zone	Residential Zone	[(40.730610, -73.935242), (40.729610, -73.934242)]	Residential Zone

iii. Spatio-temporal data: It combines both spatial (location-based) and temporal (time-based) aspects, providing how a phenomenon changes over time and across various geographical locations.
- Quering and indexing is not really easy, special techniques are required.
E.g. GPS tracking data.

Similarity measures

In data mining, we need some methods to quantify how similar 2 objects or datasets are. These measures are essential for tasks like:

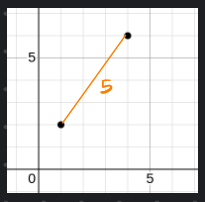
a. clustering: grouping similar items together.
b. classification: assigning labels based on similarity.
c. outlier detection: finding and removing the odd ones.

These measures help in comparing data based on their characteristics/features, enabling algorithms to make meaningful decisions or find patterns in data.

For numerical data

Euclidean distance: It is the straight line distance between 2 points.

$$d(p,q) = \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}$$

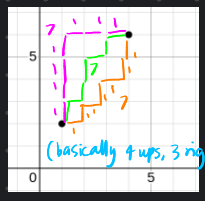


e.g. $p=(1,2)$, $q=(4,6)$

$d(p,q) = \sqrt{(4-1)^2+(6-2)^2} = 5$

Manhattan distance (L1/ city block distance): It is the sum of absolute differences of the coordinates of the pointss.

$$d(p,q) = \sum_{i=1}^{n}|p_i - q_i|$$



e.g. $p=(1,2)$ $q=(4,6)$

$d(p,q) = |4-1| + |6-2| = 7$

(basically 4 ups, 3 rights in any arrangement)

Minkowski distance: It is a generalization of the above.

$$d(p,q) = (\sum_{i=1}^{n}|p_i - q_i|^k)^{\frac{1}{k}}$$

k=1: Manhattan distance
k=2: Minkowski distance

Example on k-means clustering

Consider an example where we want to group 5 points into 2 clusters using k-means.

point	x	y
A	1	2
B	2	3
C	3	4
D	8	8
E	9	9

Choose 2 data points randomly, say A and D. We'll assign all the points to either A's or D's cluster.

point	distance from A(1, 2)	distance from D(8, 8)	cluster
A	0	8.49	A
B	1.41	7.21	A
C	2.83	5.83	A
D	8.49	0	D
E	10	1.41	D

Cluster assignment: {A, B, C} and {D, E}.

Recalculate the centroids of both the clusters.
Cluster 1 {A, B, C}: centroid = (2, 3)
Cluster 2 {D, E}: centroid = (8.5, 8.5)

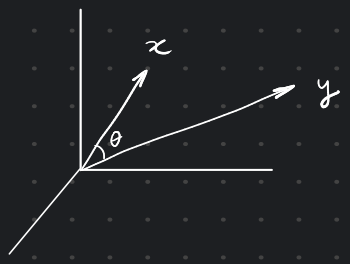
point	distance from C1(2, 3)	distance from C2(8.5, 8.5)	cluster
A	1.41	9.19	C1
B	0	8.49	C1
C	1.41	7.07	C1
D	8.49	0.707	C2
E	9.19	1.41	C2

The algo converges here, so we stop.
Clusters are {A, B, C} and {D, E}.

Cluster assignment: {A, B, C} and {D, E}.

For non-numerical data

Cosine similarity: It measures the similarity between 2 vectors of an inner product space. It is measured by the cosine of the angle between the 2 vectors.
We use this when we are more concerned with the orientation of the vectors as opposed to the shortest length between them.
The cosine of angles between 2 vectors indicates how similar the directions of the 2 vectors are.



$$\text{cosine_similarity}(x, y) = \frac{x \cdot y}{||x||||y||}$$

(goes from -1 to 1)

Q. consider 2 vectors:
$x = (5,0,3,0,2,0,0,2,0,0)$ How similar are the vectors?
$y = (3,0,2,0,1,1,0,1,0,1)$

→ $||x|| = \sqrt{42}$, $||y|| = \sqrt{17}$

$\cos\theta = \frac{15+6+2+2}{\sqrt{42}\sqrt{17}} = \frac{25}{\sqrt{42}\sqrt{17}} = 0.9356$ Really similar.

Consider a text example:

Statement 1: data mining finds patterns.
Statement 2: pattern recognition finds patterns.

We'll first tokenize the statement into words and then form term frequency vectors.

Statement 1: {'data', 'mining', 'finds', 'patterns'}
Statement 2: {'pattern', 'recognition', 'finds', 'patterns'}

Since we need to calculate the inner product, both the vectors ought to have the same dimension (here, the size of the term freq vector).

Combining these 2 sets to form a tf-vector:
tf-vector: {'data', 'mining', 'finds', 'patterns', 'pattern', 'recognition'} size = 6

tf-vector forms:
Statement 1: a = (1, 1, 1, 1, 0, 0)
Statement 2: b = (0, 0, 1, 1, 1, 1)

$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{||\vec{a}||\,||\vec{b}||} = \frac{2}{\sqrt{4} \cdot \sqrt{4}} = \frac{1}{2} = 0.5$

If both the statements are equal, the cosine would be 1. On the other hand if they are totally different (the vectors would be something like 00001111 and 11110000, no common tokens and the inner product would be 0), the similarity would be zero.

Jaccart similarity: It measures the similarity between 2 sets by comparing the size of their intersection with the size of their union.

$$\text{Jaccart_similarity} = \frac{|A \cap B|}{|A \cup B|}$$

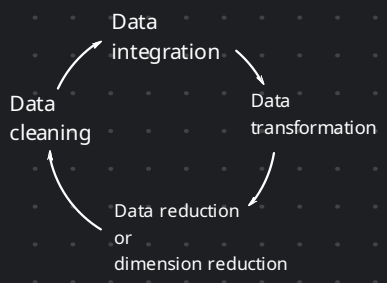
Consider the statements:
1: "data mining is the process of discovering patterns"
2: "pattern recognition and data mining find patterns"

$|A \cap B| = |\{ \text{data, mining, patterns} \}| = 3$
$|A \cup B| = |\{ \text{data, mining, is, the, process, of, discovering, patterns, pattern, recognition, and, find} \}| = 12$

$$\text{Jaccart_similarity} = \frac{3}{12} = 0.25$$

Data pre-processing

It is the process of transforming raw data into a format that is more suitable for decision making and analyzing.



Data cleaning:

It is the removal of inaccurate, incomplete, and incorrect data attributes that might be present in the dataset.

OrderID	CustomerID	Product	Quantity	Price	OrderDate	ShippingDate
1001	C001	Laptop	1	899.99	2024-06-15	2024-06-17
1002	C002	Smartphone	2	499.99	2024-07-01	2024-07-03
1003	C003	Headphones		199.99	2024/07/02	2024-07-05
1004	C004	Mouse	2	29.99	2024-07-02	2024-07-07
1005	C001	Laptop	1	899.99	2024-06-15	2024-06-17
1006	C005	Keyboard	1	49.99	2024-07-10	2024-07-10
1007	C006	Laptop	1	1199.99	2024-07-06	2024-07-08
1008	C007	Mouse	2	29.99	2024-07-10	2024-07-12
1009	C003	Headphones	1	199.99	2024-07-02	2024-07-05
1010	C008	Smartphone	1	499.99	2024-07-12	2024-07-15
1011	C009	Laptop	1	899.99	2024-07-12	2024-07-20
1012	C010	Monitor	1	349.99	2024-07-15	2024-07-18
1013	C011	Mouse		29.99	2024-07-16	2024-07-20
1014	C002	Smartphone	2	499.99	2024-07-01	2024-07-03
1015	C012	Keyboard	1	49.99	2024-07-18	2024-07-20
1016	C013	Laptop	1	999.99	2024-07-20	2024-07-25

outlier

missing

Issues with the data:
- missing values
- inconsistent data representation (date format)
- outliers (prices)
- duplicate records

a. Missing data:

Ways to handle:
- delete the record: discarding the records with missing values
- imputation: filling the missing values either by replacing them with the mean, mode, or median of the other values present in the dataset, or we try to predict the values based on the other attributes.

CustomerID	Name	Age	Email	PurchaseAmount
1	John Doe	30	john.doe@example.com	250.50
2	Jane Smith		jane.smith@example.com	
3	Jim Brown	45		180.00
4		28	mary.jane@example.com	120.75
5	Tim Clark	33	tim.clark@example.com	350.00

could use mean (34)

(225.31)

b. Noisy data:

Noise is random error or variance in a measured variable. We need to smooth out the data to remove the noise.

i. Binning: This method smoothens a sorted data value by consulting its neighbors.

E.g. consider this sorted price data: 4, 8, 15, 21, 21, 24, 25, 28, 34.

We'll partition it into bins (here equal-sized).

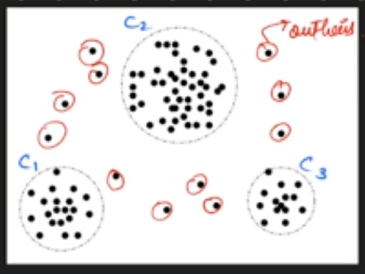
Bin 1	Bin 2	Bin 3
4, 8, 15	21, 21, 24	25, 28, 34

Smoothing by mean:

Bin 1	Bin 2	Bin 3
9, 9, 9	22, 22, 22	29, 29, 29

Smoothing by bin boundaries:

Bin 1	Bin 2	Bin 3
4, 4, 15	21, 21, 24	25, 25, 34



ii. Outlier analysis:

Clustering is widely used to detect outliers.

So the values outside the clusters can be labelled as outliers or noisy values and can be removed.

The inter-quartile range could be a measure to check for outliers.

Data points outside (Q2 ± 1.5 IQR) can be discarded as outliers.

iii. Regression: Since the output in case of regression is a continuous function, we can estimate the noisy attribute.

Data integration:

It is the combining of data from multiple sources (could be across various databases, files, etc.).

Customer ID	Name	Email	Phone
1	John Doe	john.doe@example.com	555-1234
2	Jane Smith	jane.smith@example.com	555-5678

Sale ID	Customer ID	Product	Amount	Date
101	1	Laptop	1200	2024-07-10
102	2	Smartphone	800	2024-07-11
103	1	Keyboard	100	2024-07-12



Customer ID	Name	Email	Phone	Sale ID	Product	Amount	Date
1	John Doe	john.doe@example.com	555-1234	101	Laptop	1200	2024-07-10
1	John Doe	john.doe@example.com	555-1234	103	Keyboard	100	2024-07-12
2	Jane Smith	jane.smith@example.com	555-5678	102	Smartphone	800	2024-07-11

Main challenges:

- data heterogenity: data can be from differently formatted sources.
- quality and scalability: handling large volumes of data can be complicated.
- timelessness: ensuring integrated data is up-to-date and accurate.

Data transformation

It is the process of converting data from its original (raw) format into a format suitable for data mining.
Techniques: normalization, discretization, sampling, and compression.
Purpose:
- To maintain consistency/ uniformity in data.
- To improve the quality of predictions.
- To ease the data integration process.

Normalization

It is the process of transforming data into a common scale without distorting differences in the range of values (scaling numerical data to some standard range like [-1, 1] or [0, 1]).

Why does data need to be normalized?
- To handle different scales.
  Some algos use distance metrics. These can be heavily influenced by difference in scale.
- Ease of interpretation or visualization.
- To improve rate of convergence in gradient descent.

Min-max normalization

It scales a feature to a fixed range [0, 1].

$$x' = \frac{x - min(x)}{max(x) - min(x)}$$

Employee	Age	Salary	Years of Experience
A	25	50	2
B	35	80	8
C	45	120	15
D	50	200	20
E	28	60	5

min = 25, max = 50 | max = 200, min = 50 | max = 20, min = 2

$age' = \frac{age - 25}{25}$

age'	sal' = $\frac{sal - 50}{150}$	exp'
0	0	0
0.4	0.2	0.33
0.8	0.47	0.72
1	1	1
0.12	0.07	0.17

Q. Normalize the attributes using min-max normalization, range = [-1, 1].

House	Price (in $1000s)	Size (sq ft)	Age (years)
A	500	2000	10
B	700	2500	5
C	600	1800	15
D	800	3000	20
E	550	2200	8

min = 500, max = 800

$Price' = \left(\frac{price - 500}{300}\right) \cdot 2 - 1$

0, 2/3, 1/3, 1, 1/6 → [0,1] → ×2 → 0, 4/3, 2/3, 2, 1/3 → [0,2] → −1 → −1, 1/3, −1/3, 1, −2/3 → [−1,1]

$$x' = \frac{x - min(x)}{max(x) - min(x)} \cdot (b - a) + a$$

for scaling into range [a, b]

Z-score normalization

(standardization)

It transforms data to have a mean of 0 and std dev of 1.

$$z = \frac{x - \mu}{\sigma}$$

Student	Exam Score	Hours Studied	Age
A	85	5	20
B	90	6	21
C	78	4	19
D	92	7	22
E	88	5	20

$\mu = \frac{\sum x_i}{n} = 86.6$ $\quad \mu = 5.5$ $\quad \sigma = 1.14$ $\quad \mu = 20.4$ $\quad \sigma = 1.14$

$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})}{n}} = 5.08$

ES' $= \frac{ES - 86.6}{5.08}$

ES'	h'	age'
-0.32	-0.35	-0.35
0.67	0.53	0.53
-1.69	-1.23	-1.23
1.06	1.41	1.41
0.28	-0.35	-0.35

Student	Math Score	Physics Score	Chemistry Score
A	85	60	78
B	90	70	88
C	78	55	95
D	92	65	65
E	88	80	80

$\sigma = 4.88$ ← considered popⁿ std. devⁿ.

$\mu = 86.66$

Math score (new)

-0.34
0.68
-1.77
1.09
0.27

NOTE: If σ is replaced by mean absolute deviation, $S_A = \frac{1}{n} \sum |x_i - \bar{x}|$ $\quad z' = \frac{x_i - \bar{x}}{S_A}$

Decimal scaling normalization

It is normalization done by shifting the decimal point.

Steps:

- Identify the max abs. value.
- Find the scaling factor.
- Divide the values by the scaling factor.

$$x'_i = \frac{x_i}{10^j}$$

j : smallest integer s.t. $\max(|x_i|^j) < 1$

E.g. 205, 430, 120, 345, 260 → 0.205, 0.430, 0.120, 0.345, 0.260

max value = 430

scaling factor = 1000

Q. Normalize the same to [-1, 1].

→ $\frac{x_i - 120}{430 - 120} \times 2 - 1$ → -0.451 | -1 | 0.451 | -0.096

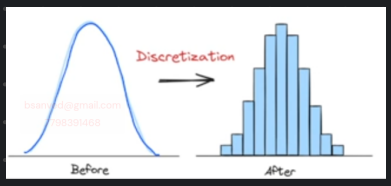
Log normalization $x' = \log x$

Unit vector normalization $x' = \frac{x}{||x||}$

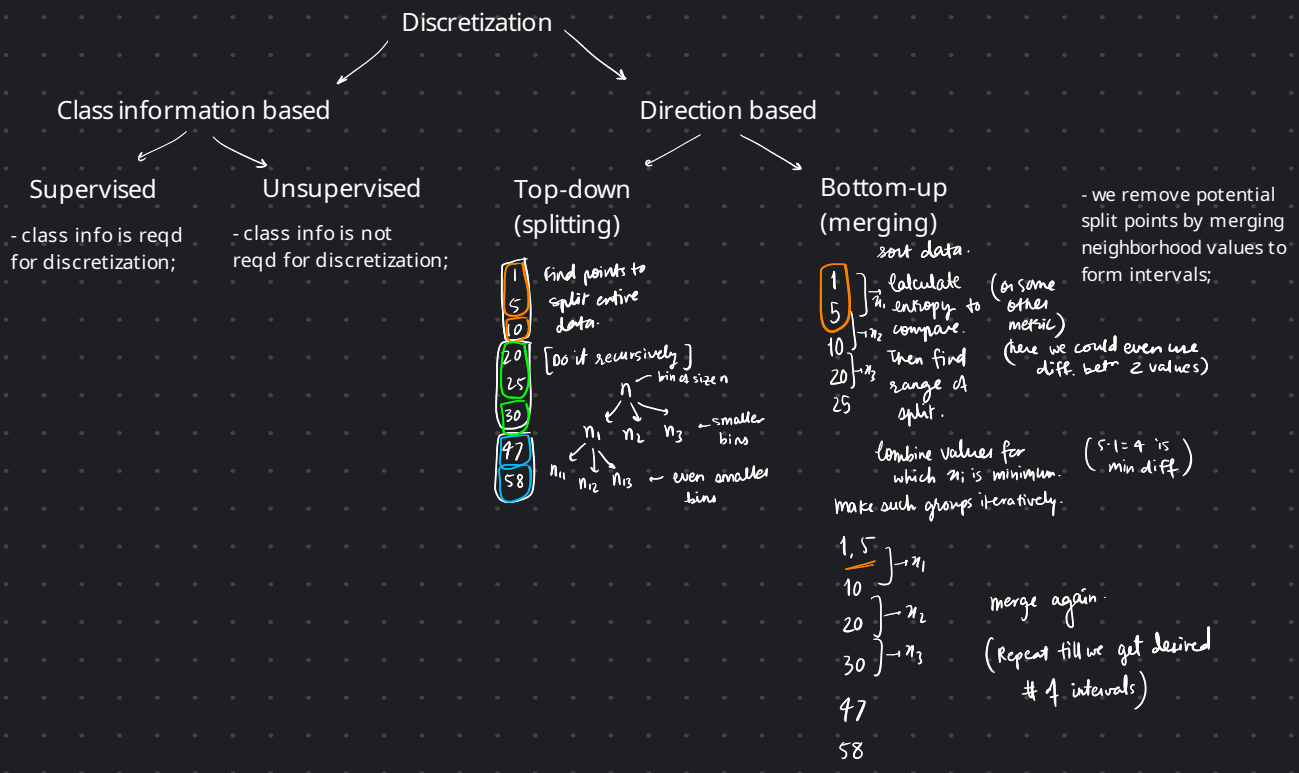
Technique	Pros	Cons
Max-min normalization	- simple to implement and understand; - rescales the value based on the variability the data have; (relative diff stays same) - rescaling to a fixed range;	- sensitive to outliers (as they'll increase the diff between max and min; - if a new point falls outside the original range, the transformation becomes invalid;
Z-score	- handles outliers better than max-min by centering the data around the mean;	- prior knowledge of mean and std dev is reqd; - transformed data may not be easily interpretable;
Decimal scaling	- really easy to implement; - useful for handling values in a large range by scaling down;	- less effective;

Discretization

It is the process of converting continuous data into discrete intervals (bins).



Method:
Using some technique, identify the bins for all the data points.
Once the groups are identified, assign labels to all the data points. These labels are usually ordinal values (ranking is maintained).



Concept heirarchy

Binning

Q. Consider the dataset: age: {23, 35, 42, 27, 51, 29, 45, 33, 39, 48}. Find the bin width and split the data into 5 equal sized bins.

→ sorted values : {23, 27, 29, 33, 35, 39, 42, 45, 48, 51}
↑ min ↑ max

Bin width = $\frac{51-23}{5} = 5.6$

Bin 1 : [23 – 28.6) : {23, 27}
Bin 2 : [28.6 – 34.2) : {29, 33}
Bin 3 : [34.2 – 39.8) : {35, 39}
Bin 4 : [39.8 – 45.4) : {42, 45}
Bin 5 : [45.4 – 51) : {48, 51}

{23, 27, 29, 33, 35, 39, 42, 45, 48, 51}
bin 1 bin 2 bin 3 bin 4 bin 5

These are both equal-width and equal frequency bins.

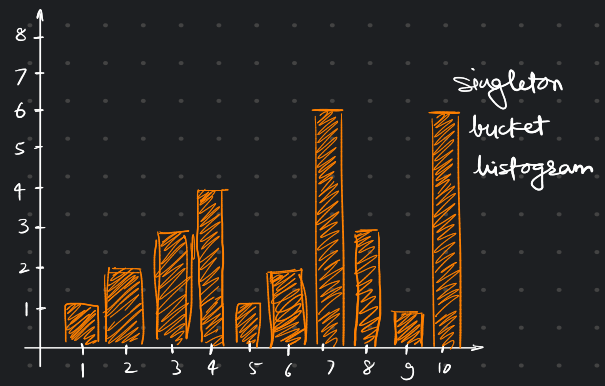
NOTE: Binning is
- top-down
- unsupervised

Bins can be
- equal width
- equal frequency

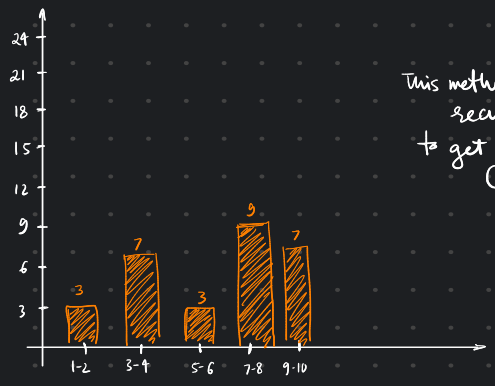
Histogram analysis

- Dividing the data into bins and then computing the frequency of observations for each bin.

{1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 6, 6, 7, 7, 7, 7, 7, 8, 8, 8, 9, 10, 10, 10, 10, 10, 10}
1 2 3 4 1 2 6 3 1 6



singleton bucket histogram



This method can be applied recursively to get coarser bins. (or less granular)

Data compression

In these techniques, the transformations are applied so as to obtain a reduced or compressed representation of the original data.

Why compression?

- storage efficiency ++, cost of storage --
- faster data transmission
- increased performance (less train time)

$$\text{compression ratio} = \frac{\text{size of original data}}{\text{size of compressed data}}$$

Compression

Lossless

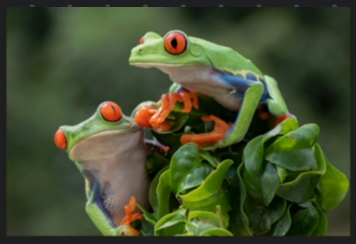
- Reconstruction of the original data back from the compressed data can be done without loss of information.

original data D_1 → compressed data → original form D_2

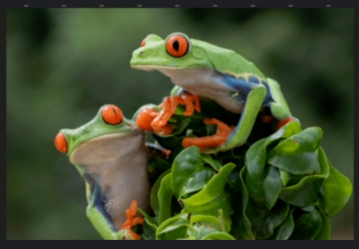
$D_1 = D_2$

(reconstruction is perfect)

- Mostly used when data integrity and precision are essential. E.g. financial data.



4.69 MB (original image)



2.82 MB (compressed image)
(there are some distortions in the image, but it's okay if we aren't losing much useful information)

- E.g. compressing to .zip files is lossless.

Lossy

- Reconstruction from compressed data would be similar to the original, but there'll be loss of information (the new data could be a subset of the original).

original data D_1 → compressed data → original form D_2

$D_1 \approx D_2$ or $D_1 \subset D_2$

(reconstruction is an approximation)

Lossless compression techniques → Huffman coding

Huffman coding

E.g. A B A B A C A D A] → we'd need to transmit $9B = 9 \times 8 = 72$ bits.

Step 1: Create a character frequency table, sorted by frequency.

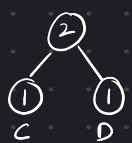
character	frequency
A	5
B	2
C	1
D	1

(1, C), (1, D), (2, B), (5, A) } freq-tuples (sorted)

Step 2: Generate a Huffman tree

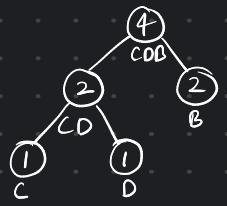
Take 2 leaves (tuples) at a time. The lower freq. on the LHS.

(1, C) (1, D)



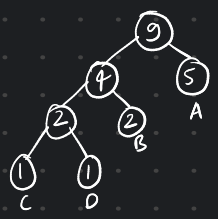
↑ add frequencies

new order: (2, CD), (2, B), (5, A)



(4, CDB)

(4, CDB), (5, A)

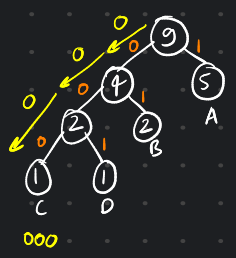


(9, CDBA)

We terminate when we reach a single element.

Step 3: Give encoding to branches

Rule:
left branch: 0
right branch: 1



Character	Frequency	Encoding
A	5	1
B	2	01
C	1	000
D	1	001

← traversal from parent

Size of table
= 4×8 + 9
(4×8: for chars; 9: encoded words)
= 41 bits

Step 4: Convert the message to encoded form

A B A B A C A D A
[1 01 1 01 1 000 1 001 1] → 15 bits ← for encoded message

We need to send these 15 bits + the encoding table.

Size of transmission = size of encoded message (15) + size of encoding table (41)
= 56 bits (as opposed to 72 bits)

Decoding

We get table + message

Character	Frequency	Encoding
A	5	1
B	2	01
C	1	000
D	1	001

start with the first bit and go ahead till we get some encoded word.

1 0 1 1 0 1 1 0 0 0 1 0 0 1 1

1 → A

1 0 1 1 0 1 1 0 0 0 1 0 0 1 1
A ← Nothing
→ B

1 0 1 1 0 1 1 0 0 0 1 0 0 1 1
A B 1 → A, 01 → B

1 0 1 1 0 1 1 0 0 0 1 0 0 1 1
A B A B A 000 → C

1 0 1 1 0 1 1 0 0 0 1 0 0 1 1
A B A B A C A 001 → D A

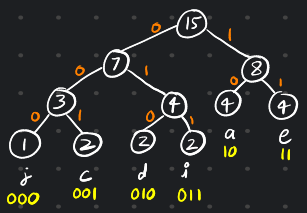
Decoded message = ABABACADA

NOTE: No code will be the prefix of some other code. So there's no ambiguity.

Q. Encode the message using Huffman coding: aaacdacdeeeeiij

→ (j,1), (i,2), (c,2), (d,2), (e,4), (a,4)

(1,j), (2,c), (2,d), (2,i), (a,4), (e,4)



a a a c d a c d e e e e i i j → 15 × 8 = 120 bits

1 0 1 0 1 0 0 0 1 0 1 0 1 0 0 0 1 0 1 0 1 1 1 1 1 1 1 1 0 1 1 0 1 1 0 0 0 → 37 bits + (6×8 + 16 =) 64 for table

= 101 for transmission

Run-length encoding (RLE)

Runs of data are stored as a single data value with the count.

E.g. 1,1,1,1,1 (5), 2,2 (2), 3,3,3 (3), 4,4,4,4,4,4 (6), 5,5 (2), 6,6,6,6,6 (5)

5,1, 2,2, 3,6, 2,5 } encoded data

Only effective if the data has many such runs. So not really effective.

Lempel-Ziv-Welch (LZW) encoding

- Dictionary-based encoding: we build a dynamic dictionary.

Steps:

- Initialization: the dictionary is initialized to single characters in the message.
- Building the dictionary: adding new multi-character sequences to the dictionary.
- Get codes: Check the dictionary and get the encoded message.

E.g. abababab.

Initial dictionary

encoding	index	entry
	1	a
	2	b

a b ab ab ab

↑ start next traversal here
make new entry

encoding	index	entry
	1	a
	2	b
	3	ab

ab ab ab ab

+ start next

encoding	index	entry
	1	a
	2	b
	3	ab
	4	ba

aba ba ba b

+

encoding	index	entry
	1	a
	2	b
	3	ab
	4	ba
	5	aba

a b a b a b a b

+

encoding	index	entry
	1	a
	2	b
	3	ab
	4	ba
	5	aba
	6	abab

At each step, start traversal from the end of the last added part, and add the next part that is not present in the dictionary.

encoding	index	entry
———	1	a
———	2	b
1	3	ab
2	4	ba
3	5	aba
5	6	abab

abab abab

1 2 3 5 2 ← Encoded message

For encoding, we start with the message abababab.
We'll start with first character. If element is present in dict, we'll increase length of subsequence to check.

ab ab ab ab

present

not present

give 1 (index of a)

start again with last char added.

ab ab ab ab

give 2 (index of matching entry of b)

1 start here

a b a b ab ab

give 3 (index of ab)

a b a b a b a b

give 5 (index of aba)

Message: 1 2 3 5 2 ← encoding of last char

Q. Message: ababbabcababba. Give the LZW encoding.

→

encoding	index	entry
	1	a
	2	b
	3	c
1	4	ab
2	5	ba
4	6	abb
5	7	bab
2	8	bc
3	9	ca
4	10	aba
6	11	abba

Index of ab

Index(ba)

a b a b b a b c a b a b b a

1 2 3 4 5 6 7 8 9 10 11

not present

124523461 ← of last char.

Decoding an LZW encoding

124523461, initialized dict

a	1
b	2
c	3

Index	entry
1	a
2	b
3	c
4	ab

We'll encounter

default entry → Put new entry prev string + single char

entry made by us → Put new entry as prev string + 1st new char encountered

Index	entry
1	a
2	b
3	c
4	ab

1 2 4 5 2 3 4 6 1

a b

entry ab

Index	entry
1	a
2	b
3	c
4	ab
5	ba

1 2 4 5 2 3 4 6 1

a b ab

new entry ba

Index	entry
1	a
2	b
3	c
4	ab
5	ba
6	abb

1 2 4 5 2 3 4 6 1

a b ab ba

new entry abb

Index	entry
1	a
2	b
3	c
4	ab
5	ba
6	abb
7	bab
8	bc
9	ca
10	aba
11	abba

1 2 4 5 2 3 4 6 1
a b ab ba b c ab abb a
entry bab bc ca aba abba

↙ same dict as previously constructed.

Decoded message : ab ab ba b c ab abb a

Q. Consider the LZW decoded message: 12352. {a:1, b:2}.

index	char
1	a
2	b
3	ab
4	ba

1 2 3 5 2 (special case)
a b ab
ab ba

case 3:
we've encountered a code not yet added to the dictionary.
If the corresponding entry is not found, add new entry
prev string + 1st char of prev string

index	char
1	a
2	b
3	ab
4	ba
5	aba

1 2 3 5 2
a b ab
ab ba
ab a

index	char
1	a
2	b
3	ab
4	ba
5	aba
6	abab

1 2 3 5 2
a b ab aba b
ab ba abab

message : ab ab aba b

Q. Encode abcaaabb.
→

encoding	index	entry
	1	a
	2	b
	3	c
1	4	ab
2	5	bc
3	6	ca
1	7	aa
7	8	aab
2	9	bb

a b c a a a b b

encoded: 1231722

Q. aabbcabcca

encoding	index	char
	1	a
	2	b
	3	c
1	4	aa
1	5	ab
2	6	bb
2	7	bc
3	8	ca
5	9	abc
3	10	cc

aa bb ca bcca ***

11223538 ← for ca

We add the left out characters at the end.

For decoding:

11223531 abc

index	char
1	a
2	b
3	c
4	aa
5	ab
6	bb
7	bc
8	ca
9	abc
10	cc

1 1 2 2 3 5 3 8
a a b b c ab c ca
aa ab bb bc ca abc cc

Q 1231722 decode abc

→

1	a
2	b
3	c
4	ab
5	bc
6	ca
7	aa
8	aab
9	bb

1 2 3 1 7 2 2
a b c a aa b b
ab aa

Lossy compression techniques

These data compression techniques reduce the file size by reducing the data.

$$\text{compression ratio} \propto \frac{1}{\text{quality of reconstruction}}$$

- Data loss: some original data may get lost.
- Higher CR: higher CR compared to lossless techniques.
- Commonly used for multimedia compression.

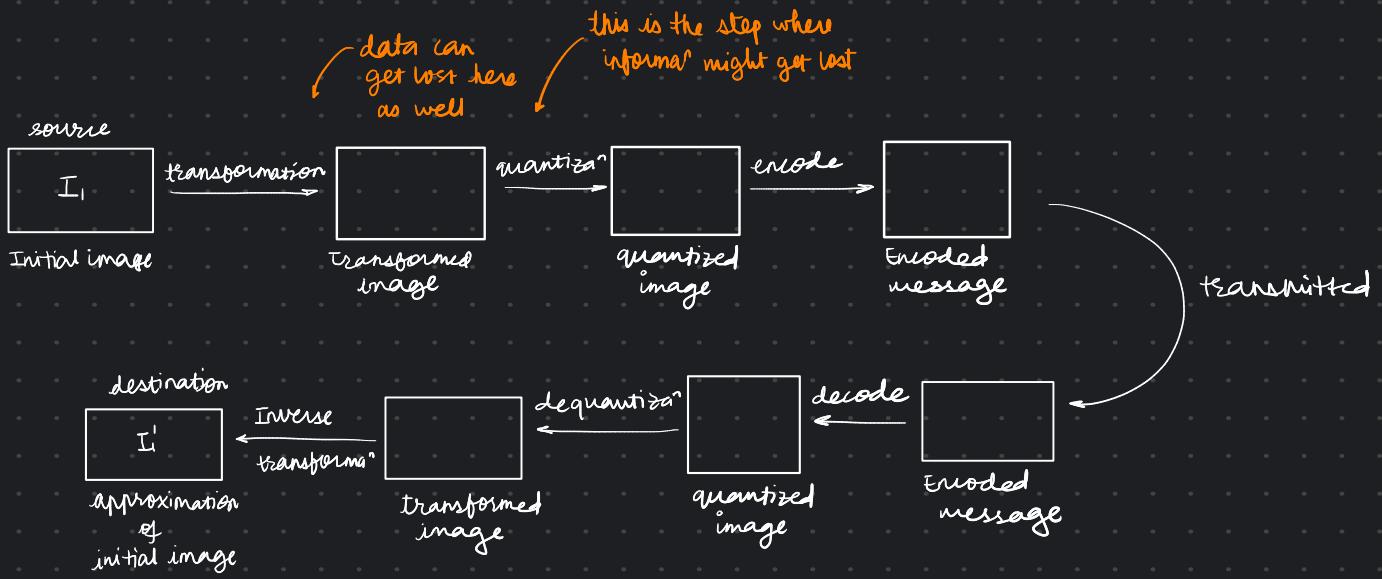


Low-compression (high quality) JPEG

High-compression (low quality) JPEG

Transform coding

1. Transformation: The original data is converted to a new domain using some mathematical functions.
2. Quantization: The transformed data is quantized to reduce precision.
3. Encoding: Just read the quantized data and encode it (using lossless techniques). Mostly running-length encoding and Huffman encoding is used to transform the quantized data.



E.g. lets consider a 4x4 block of pixels in an image.

$$x = \begin{bmatrix} 52 & 55 & 61 & 66 \\ 73 & 78 & 87 & 85 \\ 101 & 103 & 106 & 107 \\ 122 & 121 & 111 & 100 \end{bmatrix}$$

A colored img has RGB components.



All steps are applied to each.

Lets consider a simplified example.

Step 1: transforma"

We'll apply discrete cosine transformation (DCT).

$$x(u,v) = \alpha(u)\cdot\alpha(v)\sum_{x=0}^{n-1}\sum_{y=0}^{m-1} f(x,y)\cdot\cos\left[\frac{(2x+1)u\pi}{2n}\right]\cdot\cos\left[\frac{(2y+1)v\pi}{2m}\right]$$

$x(u,v)$: DCT coefficient or transformed value

where $\alpha(u) = \sqrt{\frac{1}{n}}$ if $u=0$; $\sqrt{\frac{2}{n}}$ if $u>0$

$f(x,y)$ = pixel value in matrix $[x][y]$

$$\begin{bmatrix} 52 & 55 & 61 & 66 \\ 73 & 78 & 87 & 85 \\ 101 & 103 & 106 & 107 \\ 122 & 121 & 111 & 100 \end{bmatrix}$$

$$x(0,0) = \alpha(0)\cdot\alpha(0)\sum_{x=0}^{3}\sum_{y=0}^{3} f(x,y) \qquad \text{as } \cos 0 = 1$$

$$= \sqrt{\tfrac{1}{4}}\sqrt{\tfrac{1}{4}} \times (1219)$$

$$= 304.75$$

Transformed matrix:

$$\begin{bmatrix} 304.75 & -17 & 0 & 0 \\ -90 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Any significant data would lie in a small section of the transformed matrix.

Step 2 : Quantizaⁿ

We can either take a scalar (SF) factor and divide all the elements by it.

Say $SF = \frac{1}{10}$

$\frac{1}{10} \times TF$ = Quantized matrix =

$$\begin{bmatrix} 34 & -2 & 0 & 0 \\ -4 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

-1.7 → -2

we round-off as well so there's irrecoverable data loss.

$$\begin{bmatrix} 16 & 11 & 10 & 16 \\ 12 & 12 & 14 & 19 \\ 14 & 13 & 16 & 24 \\ 14 & 17 & 22 & 29 \end{bmatrix}$$

scaling matrix (random mxn grid)

We could also use a scaling matrix. But that needs to be remembered for de-quantizaⁿ as well.

$$\hookrightarrow Qm = \begin{bmatrix} 21 & -2 & 0 & 0 \\ -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

we get Quantized DCT coefficients.

Now before encoding, we take a zig-zag scan order.

$$\begin{bmatrix} 21 & -2 & 0 & 0 \\ -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

After a certain pt, all values are 0 so encoding becomes easier.

Values: [21, -2, -3, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]

(usually transformations give values like these)

Step 3 : Encoding

RLE and/or Huffman encoding is used.

RLE = {1, 21, 1, -2, 1, -3, 13, 0}

JPEG compression

Steps:

1. Loading the RGB image.
2. Convert the image to the YCbCr color space. Y: Luma component (luminance), Cb: blue-difference, Cr: red-difference;
   It's observed that the human eye is more sensitive to brightness (Y) than it is to color.
   So we'll transmit the Y component without much hampering. The Cb and Cr components will be sent in low-res.
3. For each channel (dimension) (Y, Cb, Cr), we'll take the grid:
   a. Split the channel (grid) into non-overlapping 4x4 (based on performance of compression), 8x8 blocks.
   b. For each such 4x4 block,
      i. Apply DCT to convert it into frequency domain.
      ii. Perform quantization.
      iii. Apply zig-zag scan to re-order the quantized coefficients.
      iv. Encode using RLE.
      v. Apply Huffman encoding to further compress the data.
      On doing these v steps, we'll get the encoded 4x4 block. We'll aggregate all such blocks to get the whole file.
4. Output the JPEG files:
   Include metadata (Huffman tables, quantization tables, etc.).
   Add compressed image data and end-of-image markers (also separators for metadata).

Sampling

Sampling is a technique that allows analysts to work with a representative subset of data instead of the entire data set (called the population).
Advantages:
- Efficiency and speed: reduced computation load and quicker insights;
- Cost effectiveness: Low storage and handling costs;
- Improved focus and precision: Helps to achieve targeted analysis;
- Reduced chances of overfitting;

Common ways of sampling

Some common ways are simple random sampling, stratified sampling, and

Simple random sampling

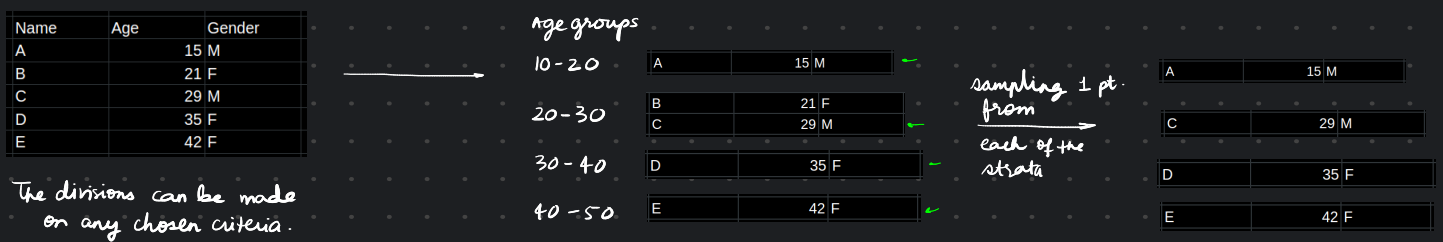
Each data point has an equal chance of being selected.



SRS
- without replacement — Everytime a sample is drawn, it is not placed back.
- with replacement — Everytime a sample is drawn, it is placed back and considered for the subsequent draws.

Stratified sampling

If the dataset D is divided into mutually disjoint parts (called strata), a stratified sample of D is generated by obtaining a sample at each strata.



It helps in ensuring a representative sample (esp. when the data is skewed).

Cluster sampling

The population is divided into clusters and a random sample of clusters is selected.



If we further take a random sample out of those clusters, it is called two-stage cluster sampling.
If we take all the elements out of those clusters, it is called cluster sampling.

Store ID	Region	Customer Feedback Score	
1	North	4.5	C_1
2	North	3.8	
3	North	4.2	
4	South	4.7	C_2
5	South	3.6	
6	South	4.1	
7	East	4.9	C_3
8	East	4.3	
9	East	3.9	
10	West	4.8	C_4
11	West	4.0	
12	West	3.7	

take all of them → cluster sampling

choose 2 clusters → C_1 (3) and C_3 (3) → sample 1 from each

8	East	4.3
1	North	4.5

(2-stage cluster sampling)

Systematic sampling

A starting point is selected at random and then every kth element in the dataset is selected.

ID	Age	Gender	Income	
1	25	M	50,000	
2	30	F	60,000	
3	22	M	45,000	← start
4	35	F	70,000	k = 2
5	28	M	55,000	←
6	33	F	65,000	
7	27	M	52,000	←

Data reduction

Methods or techniques to reduce the volume of data while maintaining its integrity and relevance for analysis.

Why?

Data sets might contain hundreds of attributes, out of which many can be irrelevant for analysis.

E.g. Whether a user will click on an ad doesn't really depend on attributes like username and phone #, but may be affected by age.

